

CSE 574 - Computational Complexity

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Abstract

This is a template for TCS (and other kind of) lecture notes.

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Part I

Introduction & Background Review

Chapter 1

The Beginning

The start of computational theory can be traced back to the 1930s, when mathematicians and logicians were trying to formalize the notion of computation and algorithms. The Church-Turing thesis, proposed independently by Alonzo Church and Alan Turing, laid the foundation for the study of computational complexity.

Theorem 1.0.1 (Church-Turing Thesis). Every effectively calculable function is computable by a Turing machine.

Then mathematicians and computer scientists began to explore the limits of computation, with such an open thesis:

Problem 1.0.1 (Extended (Quantum) Church-Turing Thesis). Every physically realizable model of computation can be simulated by a (quantum) Turing machine efficiently.

1.1 Definition of Computational Theory

Remark 1.1.1. Computational theory deals with how much resources we need to solve a computational problem.

Resources can be many things. For example:

- Time
- Memory / Space
- Samples Complexity (how many samples we need to distinguish between distributions)
- Communication Rounds
- Oracle Calls
- Randomness
- Power
- Quantum Entanglement
- Number of Parallel Workers
- ...

1.2 What is Complexity About?

In CS Theory, people do research in algorithms and complexity. People might lean on one side or the other, but they are closely related. In a word:

Remark 1.2.1. Algorithms are about the upper bound of a problem, while complexity is about the lower bound of a problem.

Chapter 2

Time and Non-deterministic Time

2.1 Languages and Problems

In computational complexity, we talk about languages as how we think about computational problems.

Definition 2.1.1. A **language** L is a set of strings that $L \subseteq \Sigma^*$, where Σ is a finite alphabet (often $\{0, 1\}$ bits.)

Example 2.1.1. An algorithm $\text{PRIMES}(n)$ can decide whether a number n is prime or not. The language of this problem is:

$$\text{PRIMES} = \{\langle x \rangle : x \text{ is prime}\}.$$

$\langle x \rangle$ is the encoding of x in binary.

2.2 Classes of Complexity

2.2.1 Deterministic Time Complexity

Definition 2.2.1. $L \in \text{DTIME}(T(n))$ if there's a deterministic Turing machine M that decides L in time $O(T(n))$, where n is the size of the input. $P := \bigcup_{c \geq 1} \text{DTIME}(n^c)$ is the class of problems that can be solved in polynomial time.

Researchers have found a polynomial time algorithm for the PRIMES problem in 2002, so $\text{PRIMES} \in P$.

Example 2.2.1.

$$\text{CONN} = \{x : x \text{ is the adjacency matrix of a connected graph}\}$$

$\text{CONN} \in P$ because we can use DFS or BFS to check connectivity in polynomial time.

2.2.2 Non-deterministic Time Complexity

Definition 2.2.2. A **non-deterministic Turing machine** is a Turing machine that can make non-deterministic choices at each step. It can be thought of as a machine that can explore multiple computation paths simultaneously, or a machine that takes different branches of operations in different universes.

For instance, assume a certain state. In this state, the transition function for a deterministic TM might be just $(0, L, 1)$ and $(1, R, 1)$, while a non-deterministic TM might have $(0, R, 1)$, $(0, L, 0)$, and $(1, L, 1)$. When the machine meets a 0, it may arbitrarily choose to go left or right.

Definition 2.2.3. $L \in \text{NTIME}(T(n))$ if there's a non-deterministic Turing machine M that decides L in time $O(T(n))$ (regardless of which universe.) Here 'decides' means that if $x \in L$, there exists a computation path that accepts x ; if $x \notin L$, all computation paths reject x .

Definition 2.2.4. $L \in \text{co-NTIME}(T(n)) \Leftrightarrow \bar{L} \in \text{NTIME}(T(n))$.

Definition 2.2.5. $\text{NP} := \bigcup_{c \geq 1} \text{NTIME}(n^c)$ is the class of problems that can be solved in polynomial time by a non-deterministic Turing machine.

Remark 2.2.1. We normally consider NP to be 'verifiable' in polynomial time, using a 'certificate' u . Consider a non-deterministic TM that accepts x in a certain path. The sequence of non-deterministic choices made by the TM can be encoded as a string u . Then we can construct a deterministic TM $M(.,.)$ that takes x and u as input, and verify whether $x \in L$ by checking if there exists a certificate u such that $M(x, u)$ accepts in polynomial time. This can also explain why the certificate is required to be in size $O(T(n))$.

In mathematical terms:

$$x \in L \Leftrightarrow \exists u \in \{0, 1\}^{O(T(n))} \text{ s.t. } M(x, u) = 1.$$

Then for NP, $T(n) = \text{poly}(n)$.

Example 2.2.2.

- Independent Set.

$$\text{IND-SET} = \{\langle G, k \rangle : \exists S \subseteq V(G) \text{ s.t. } |S| \geq k \text{ and } \forall u, v \in S, \{u, v\} \notin E(G)\}$$

Independent Set is in NP because we can verify a solution by manually checking the edges in the graph.

- Traveling Salesman Problem (TSP) is about many cities, with d_{ij} being the distance between city i and city j .

The problem is whether there exists a closed circuit that visits each city exactly once, with total distance at most k .

TSP is in NP because we can calculate the distance of a circuit in polynomial time.

- Linear Programming: $A\vec{x} \leq \vec{b}$. Is there a solution $\vec{x}$?

LP is trivially NP, but people later found that LP is also in P.

- Graph Isomorphism: Given two adjacency matrices, do they define the same graph up to permutation of vertices?

Graph Isomorphism is in NP because we can verify a solution by checking if such permutation of vertices gives the same adjacency matrix.

- Factoring.

$$\text{FACTORING} = \{\langle N, k \rangle : N \text{ has a factor } \leq k\}$$

Equivalently, it can also decide $\{\langle N, L, R \rangle : N \text{ has a prime factor between } L \text{ and } R\}$.

Factoring is in NP as we can just divide N by a certificate.

Theorem 2.2.1.

$$\text{P} \in \text{NP} \cap \text{coNP}.$$

Chapter 3

Proving NP-hard Problems

3.1 NP-hard and NP-complete

Definition 3.1.1 (Karp Reduction). $L \leq_p L'$ if there's a polynomial-time computable function f such that for all x , $x \in L \Leftrightarrow f(x) \in L'$.

Definition 3.1.2 (NP-hard and NP-complete). L is **NP-hard** if for all $L' \in \text{NP}$, $L' \leq_p L$. L is **NP-complete** if L is NP-hard and $L \in \text{NP}$.

We can find a NP-complete problem by simply looking at the definitions.

Problem 3.1.1 (TMSAT).

$$\text{TMSAT} = \{ \langle M, x, 1^n, 1^t \rangle : \exists u \in \{0, 1\}^n \text{ such that } M(x, u) \text{ accepts in } t \text{ steps} \}.$$

Theorem 3.1.1. TMSAT is NP-complete.

Proof. First of all we see that TMSAT is in NP as we can give out u and simulate.

Now we want to prove that TMSAT is NP-hard. Let $L \in \text{NP}$ and let M be a polynomial-time verifier for L . Let $p(n)$ be the polynomial bound on the running time of M . We can reduce L to TMSAT by mapping x to $\langle M, x, 1^{p(|x|)}, 1^{p(|x|)} \rangle$. This reduction is clearly polynomial-time computable. Now we have that $x \in L$ if and only if there exists a u such that $M(x, u)$ accepts in $p(|x|)$ steps, which is equivalent to $\langle M, x, 1^{p(|x|)}, 1^{p(|x|)} \rangle \in \text{TMSAT}$. Therefore, TMSAT is NP-hard. ■

This, however, does not give us a very useful NP-complete problem.

3.2 Cook-Levin Theorem

Cook-Levin theorem is about an interesting NP-complete problem called SAT.

Definition 3.2.1 (CNF and SAT). A formula is in **conjunctive normal form** (CNF) if it is a Boolean formula over variables $u_1, u_2, \dots, u_n$ that has the form

$$\bigwedge_i \left(\bigvee_j v_{ij} \right)$$

where each v_{ij} is either a variable u_k or its negation $\neg u_k$ (also written as $\overline{u_k}$). A formula is **satisfiable** if there exists an assignment of truth values to the variables that makes the formula evaluate to true. The **SAT** problem is the decision problem of determining whether a given CNF formula is satisfiable.

Theorem 3.2.1 (Cook-Levin Theorem). SAT is NP-complete.

Proof. It suffices to show that SAT is NP-hard.

Let $L \in \text{NP}$. By definition, there exists a polynomial-time verifier M for L . We want to describe a polynomial-time transformation $x \rightarrow \varphi_x$ such that

$$\varphi_x \in \text{SAT} \Leftrightarrow \exists u \in \{0, 1\}^{p(|x|)} \text{ such that } M(x, u) = 1$$

We do so by constructing a tableau of the computation of M on input (x, u) , where u is a string of length $p(|x|)$. The tableau is a two-dimensional array where each row represents a configuration of the Turing machine at a specific time step, and each column represents a tape cell.

The first row corresponds to the initial configuration. For example, for a starting state q_0 and input tape containing x followed by u , the first row would look like:

$$\perp \quad \underbrace{q_0}_{\text{state}} \quad \underbrace{w_1 w_2 \dots w_n}_{\text{input tape}} \quad \underbrace{\#\#\dots}_{\text{blank cells}} \quad \perp$$

Checking the first row is easy, as we can hardcode the initial configuration into the formula. Name such a formula φ_{start} .

Then after some state transitions, the intermediate rows may look like:

$$\perp \quad \underbrace{\#\#\dots}_{\text{blank cells}} \quad \underbrace{y_1 y_2 \dots y_i}_{\text{tape contents}} \quad \underbrace{q_j}_{\text{state}} \quad \underbrace{y_{i+1} \dots y_m}_{\text{tape contents}} \quad \underbrace{\#\#\dots}_{\text{blank cells}} \quad \perp$$

We can't be sure about the contents of the tape, but we can check that the transition from one row to the next is valid according to the transition function of M . For example, describing a transition on state q_i only concerns a 2×3 window of the tableau:

$$\begin{array}{ccc} \# & q_i & 0 \\ q_j & \# & 1 \end{array}$$

Then it suffices to encode the transition rules of M into a windows, and check that every 2×3 window of the tableau satisfies one of the transition rules. This can be done by a formula φ_{move} .

Also, we want to check that all cells are valid (i.e. there're no multiple characters in a cell) We encode such constraints into a formula φ_{cell} .

Finally, there has to be a q_{accept} in the last row of the tableau. Searching for this symbol can be encoded into a formula φ_{accept} .

Now we construct the final formula as $\varphi_x = \varphi_{\text{start}} \wedge \varphi_{\text{move}} \wedge \varphi_{\text{cell}} \wedge \varphi_{\text{accept}}$. This formula is satisfiable if and only if there exists a valid tableau of the computation of M on input (x, u) that leads to an accepting configuration, which is equivalent to saying that there exists a u such that $M(x, u) = 1$.

■

Appendix

Appendix A

Algebra

Theorem A.0.1. For any polynomial $A \in \mathbb{C}[X]$ of degree $n - 1$, any point set $\{p_0, \dots, p_{n-1}\}$ of size n , $\{(p_i, A(p_i))\}_{i=0}^{n-1}$ determines $A(x)$ and vice versa.

Theorem A.0.2 (Fundamental Theorem of Algebra). Every non-constant single-variable polynomial with complex coefficients has at least one complex root.

In math terms, for $A(x) = \sum_{i=0}^{n-1} a_i x^i$, either $\exists x \in \mathbb{C}$ such that $A(x) = 0$, or $a_0 = a_1 = \dots = a_{n-1} = 0$.

Appendix B

Analysis

Theorem B.0.1 (Master Theorem (simplified)). Let the recurrence relation be:

$$T(n) = aT\left(\frac{n}{b}\right) + n^d$$

where $a \geq 1$ is the number of subproblems, $b > 1$ is the factor by which the problem size is reduced, and $d \geq 0$. The asymptotic bound of $T(n)$ is given by:

1. If $\log_b a > d$, then $T(n) = O(n^{\log_b a})$.
2. If $\log_b a = d$, then $T(n) = O(n^d \log n)$.
3. If $\log_b a < d$, then $T(n) = O(n^d)$.

Theorem B.0.2 (Master Theorem). Let $a \geq 1$ and $b > 1$ be constants, let $f(n)$ be a function, and let $T(n)$ be defined on the nonnegative integers by the recurrence:

$$T(n) = aT(n/b) + f(n)$$

Then $T(n)$ has the following asymptotic bounds:

- **Case 1:** If $f(n) = O(n^{\log_b a - \epsilon})$ for some constant $\epsilon > 0$, then $T(n) = \Theta(n^{\log_b a})$.
- **Case 2:** If $f(n) = \Theta(n^{\log_b a} \log^k n)$ for some constant $k \geq 0$, then $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$.
- **Case 3:** If $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some constant $\epsilon > 0$, and if $af(n/b) \leq cf(n)$ for some constant $c < 1$ and all sufficiently large n (Regularity Condition), then $T(n) = \Theta(f(n))$.